

ICP Final Project Report

Project Title: Dream Vacation Planner

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1. Abstract

The Dream Vacation Planner is a basic C programming project designed to help users explore and book vacation packages. The system stores destination names, prices, and available seats using arrays. It allows users to view vacation packages, find a recommended destination based on their vacation interest, and book a trip. The program also calculates the total bill, updates the remaining seats, and keeps track of the total revenue. The project demonstrates fundamental C programming concepts such as arrays, functions, conditional statements, loops, user input, and pointers. The system is menu-driven, allowing the user to repeatedly select different operations until choosing the exit option.

2. Introduction

Planning a vacation can involve choosing a suitable destination, checking prices, and considering the number of available seats. The Dream Vacation Planner provides a simple way to organize these tasks in one program. The main objective of this project is to create a beginner-friendly vacation planning and booking system using basic C programming concepts.

The program contains four vacation destinations: Cox's Bazar, Sajek, Sylhet, and Saint Martin. Each destination has a corresponding price and number of available seats. The program allows users to view these packages, receive a destination recommendation, and book a trip according to their choice.

3. Methodology / Implementation

The program begins by declaring arrays for destination names, prices, and available seats. The destination names are stored in a two-dimensional character array, while the prices and seats are stored in integer arrays. The same array index represents the same destination across all three arrays.

A do-while loop is used in the main function to repeatedly display the program menu. The user can choose to view vacation packages, find a dream destination, book a trip, or exit the program.

The **display()** function uses a for loop to show all available vacation packages, including the package ID, destination, price, and available seats.

The **recommend()** function asks the user about their preferred type of vacation. Based on the selected option, the program recommends one of the four destinations. Conditional statements are used to match the user's choice with a destination.

The **book()** function handles the booking process. The user enters a package ID and the number of travelers. The program checks whether the package ID and number of travelers are

valid and whether enough seats are available. If the booking is successful, the total bill is calculated using the destination price and number of travelers. The available seats are then reduced, and the total revenue is updated.

The program also contains an **animation()** function. After a successful booking, it displays a moving "TRIP READY!" message using a loop, `\r`, `fflush(stdout)`, and `usleep()`.

Sample Code Visual – Booking

```
total = price[id] * people;

seats[id] = seats[id] - people;

*revenue = *revenue + total;

printf("\n===== \n");
printf("    BOOKING SUCCESSFUL! \n");
printf("===== \n");
printf("Destination: %s \n", destination[id]);
printf("Travelers: %d \n", people);
printf("Total Bill: %d Taka \n", total);
printf("Remaining Seats: %d \n", seats[id]);
```

Figure 1. Example code used for calculating and displaying a successful vacation booking.

4. Results / Output

The completed program provides a simple menu through which the user can explore vacation packages, receive a destination recommendation, and book a trip.

In a sample run, the user can view the four available destinations and then select a package for booking. After entering the number of travelers, the program calculates the total bill and updates the remaining seats.

Sample Input / Output

```
=====
    DREAM VACATION PLANNER
=====
1. View Vacation Packages
2. Find My Dream Destination
3. Book Trip
4. Exit
=====
```

Enter your choice: 3

Package ID: 1

Destination: Cox's Bazar

Price: 5000 Taka/person

Available Seats: 20

Package ID: 2

Destination: Sajek

Price: 7000 Taka/person

Available Seats: 15

Package ID: 3

Destination: Sylhet

Price: 4500 Taka/person

Available Seats: 10

Package ID: 4

Destination: Saint Martin

Price: 8000 Taka/person

Available Seats: 8

Enter Package ID: 3

Enter Number of Travelers: 2

=====

BOOKING SUCCESSFUL!

=====

Destination: Sylhet

Travelers: 2

Total Bill: 9000 Taka

Remaining Seats: 8

=====

TRIP READY!

Figure 2. Sample input and output of the Dream Vacation Planner.

5. Challenges and Limitations

One challenge was keeping the destination information synchronized across the destination, price, and seats arrays. The same index must represent the correct destination in all three arrays.

Another challenge was validating user input during the booking process. The program needs to check whether the package ID is valid, whether the number of travelers is positive, and whether enough seats are available.

The program also has some limitations. It contains only four destinations and uses fixed prices and seat capacities. The program does not save booking information after it closes, and it does not provide advanced features such as changing or cancelling a booking.

6. Conclusion and Discussion

The Dream Vacation Planner successfully demonstrates how basic C programming concepts can be applied to a practical vacation planning and booking problem. The program allows users to view destinations, receive recommendations, and book trips.

Arrays are used to organize destination information, while functions divide the program into separate tasks. Conditional statements handle user choices and validation, and the do-while loop allows the menu to be displayed repeatedly. The booking function also demonstrates the use of a pointer to update the total revenue.

7. Future Work

The system can be improved by adding more destinations and allowing users to select additional preferences such as budget, number of travelers, or type of trip. More booking options could also be added, such as cancelling or modifying a booking.

A future version could store customer names and booking details and save the information to a file. The program could also include stronger input validation and a more advanced recommendation system that considers multiple preferences before suggesting a destination.

8. References

1. Kernighan, B. W., & Ritchie, D. M. (1988). *The C Programming Language* (2nd ed.). Prentice Hall.
2. GeeksforGeeks. (n.d.). *C Programming Language Tutorial*. <https://www.geeksforgeeks.org/c/c-programming-language/>
3. Programiz. (n.d.). *C Programming*. <https://www.programiz.com/c-programming>
4. cppreference.com. (n.d.). *C Language Reference*. <https://en.cppreference.com/w/c>

9. Individual Contributions

Barsat : Contributed to the C program implementation, including the destination arrays, menu system, recommendation function, booking system, revenue calculation, testing, and preparation of the project report.

Kona : Contributed to the planning and design of the Dream Vacation Planner, testing of the program, reviewing the project flow, and preparation of the project report.

Both members: Participated in reviewing the final project, checking the program flow, identifying possible improvements, and finalizing the project.